



## CERTIFICATE OF TESTS

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Printed: 03/24/2025

Heat: 24074282  
Material: 5L- X52 4-1/2X.337 (15.00#)BR,DRL,PEB (11000102)  
Vendor: Big River Steel - Osceola AR US (180431)

Purchase Order: 26708947  
Sales Order: 1001001265  
MFG Date: 07/11/2024

### CHEMICAL PROPERTIES

|       | C          | MN     | P          | S      | NI        | CR     | CU        | MO     | SI         | TI     | V          | CE     |
|-------|------------|--------|------------|--------|-----------|--------|-----------|--------|------------|--------|------------|--------|
| Ladle | 0.0600     | 0.5400 | 0.0140     | 0.0010 | 0.0400    | 0.1000 | 0.0800    | 0.0130 | 0.1400     | 0.0010 | 0.0020     | 0.1040 |
|       | Al: 0.0330 |        | Sn: 0.0045 |        | N: 0.0086 |        | B: 0.0003 |        | Ca: 0.0023 |        | Nb: 0.0110 |        |
| Chem  | 0.0506     | 0.5530 | 0.0092     | 0.0011 | 0.0298    | 0.0598 | 0.0746    | 0.0085 | 0.1450     | 0.0008 | 0.0030     | 0.0914 |
|       | Al: 0.0244 |        | Sn: 0.0031 |        | N: 2.0798 |        | B: 0.0001 |        | Ca: 0.0016 |        | Nb: 0.0135 |        |
| Chem  | 0.0699     | 0.5843 | 0.0096     | 0.0011 | 0.0305    | 0.0620 | 0.0776    | 0.0090 | 0.1544     | 0.0009 | 0.0033     | 0.1134 |
|       | Al: 0.0266 |        | Sn: 0.0034 |        | N: 0.1749 |        | B: 0.0001 |        | Ca: 0.0022 |        | Nb: 0.0150 |        |

Acceptance Criteria: CEPCM = 0.2500

### MECHANICAL PROPERTIES

YS (ksi): 70.20      TS (ksi): 74.60      EL (%) : 29.00      HRB: 89.00      Y/T(%): 0.94

### CHARPY TEST RESULTS (BODY)

|                                      |             |                |        |
|--------------------------------------|-------------|----------------|--------|
| Orientation : TRANSVERSE             | Specimen(S) | Energy (ft-lb) | %Shear |
| Spec. Size (in) : .394"X.197"X2.165" | 1           | 60.00          | 100.00 |
| Test Temp (°F) : -20                 | 2           | 68.00          | 100.00 |
| Location : 3T90                      | 3           | 70.00          | 100.00 |
|                                      | Average     | 66.00          | 100.00 |

Acceptance Criteria Min. (ft-lb) : 10

### COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION  
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT  
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM  
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE  
FLATTENING TEST PASSED  
1 INCH L90 STRIP SPECIMEN  
THIS MATERIAL MEETS :ASTM A53B-24,ASME SA53B-21 LATEST EDITION  
API 5L PSL 2 GRADE X52M  
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1  
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION  
MELTED AND MANUFACTURED IN THE U.S.A.  
STEEL SUPPLIER: Big River Steel - Osceola AR US  
COIL MANUFACTURER: Big River Steel - Osceola AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.

  
Quality Control Dept